

Technical Report: RSA, Quantum Threat, and Post-Quantum Cryptography

1. The Quantum Threat

In classical RSA implementations, finding the period r of a function $a^r \equiv 1 \pmod{N}$ requires a brute-force loop that tests values one by one. This becomes infeasible for real RSA keys (2048-bit or larger), where the search space is astronomically large. Shor's Algorithm bypasses this sequential search using quantum principles. First, it places qubits into **superposition**, allowing the system to represent many possible values of r simultaneously instead of testing them individually. Next, the algorithm performs modular exponentiation across all these states in parallel. The key step is the Quantum Fourier Transform (QFT), which converts hidden periodic structure into measurable probability peaks. Through **quantum interference**, incorrect values cancel out (destructive interference), while correct periodic values are amplified (constructive interference). After measurement, the correct period r is obtained with high probability in polynomial time, avoiding exponential brute-force search.

2. The Post-Quantum Defense

Post-Quantum Cryptography (PQC) aims to build cryptographic systems resistant to both classical and quantum attacks. Unlike RSA, PQC does not rely on integer factorization or discrete logarithms. A leading approach is **lattice-based cryptography**, especially the Learning With Errors (LWE) problem. LWE gives an attacker a system of linear equations with small random noise added to each equation. Without noise, solving the system would be straightforward. However, the carefully introduced noise destroys the clean mathematical structure needed for efficient solving. This prevents both classical and quantum algorithms, including Shor's Algorithm, from exploiting periodic patterns. As a result, no known efficient quantum algorithm can solve LWE. This makes lattice-based schemes a strong candidate for secure communication in the post-quantum era. Examples include ML-KEM (Kyber) for encryption and ML-DSA (Dilithium) for digital signatures, both standardized for post-quantum security.